

Immunization against poly-*N*-acetylglucosamine reduces neutrophil activation and GVHD while sparing microbial diversity

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Edited by John J. Mekalanos, Harvard University, Boston, MA, and approved August 27, 2019 (received for review May 17, 2019)

Microbial invasion into the intestinal mucosa after allogeneic hematopoietic cell transplantation (allo-HCT) triggers neutrophil activation and requires antibiotic interventions to prevent sepsis. However, antibiotics lead to a loss of microbiota diversity, which is connected to a higher incidence of acute graft-versus-host disease (aGVHD). Antimicrobial therapies that eliminate invading bacteria and reduce neutrophil-mediated damage without reducing the diversity of the microbiota are therefore highly desirable. A potential solution would be the use of antimicrobial antibodies that target invading pathogens, ultimately leading to their elimination by innate immune cells. In a mouse model of aGVHD, we investigated the potency of active and passive immunization against the conserved microbial surface polysaccharide poly-*N*-acetylglucosamine (PNAG) that is expressed on numerous pathogens. Treatment with monoclonal or polyclonal antibodies to PNAG (anti-PNAG) or vaccination against PNAG reduced aGVHD-related mortality. Anti-PNAG treatment did not change the intestinal microbial diversity as determined by 16S ribosomal DNA sequencing. Anti-PNAG treatment reduced myeloperoxidase activation and proliferation of neutrophil granulocytes (neutrophils) in the ileum of mice developing GVHD. In vitro, anti-PNAG treatment showed high antimicrobial activity. The functional role of neutrophils was confirmed by using neutrophil-deficient *LysM^{cre} Mcl1^{fl/fl}* mice that had no survival advantage under anti-PNAG treatment. In summary, the control of invading bacteria by anti-PNAG treatment could be a novel approach to reduce the uncontrolled neutrophil activation that promotes early GVHD and opens a new avenue to interfere with aGVHD without affecting commensal intestinal microbial diversity.

microbiome | GVHD | neutrophil granulocytes

Alogeneic hematopoietic cell transplantation (allo-HCT) is a curative therapy for different malignant and nonmalignant hematological disorders. Major life-threatening complications after allo-HCT are acute graft-versus-host disease (aGVHD) and infections (1). The incidence of GVHD after allo-HCT remains high, despite prophylactic immunosuppressive medication. According to the CIBMTR (Center for International Blood and Marrow Transplant Research) database, 60% of the patients undergoing allo-HCT develop grade II to IV aGVHD and 14% develop grade III to IV aGVHD (2).

We and others reported that neutrophil granulocytes (neutrophils) infiltrate into the intestinal tract after allo-HCT, which was associated with tissue damage promoting aGVHD (3, 4).

The neutrophil-mediated tissue damage was dependent on microbial transmigration because neutrophils lacking certain pattern recognition receptors did not promote GVHD and germ-free mice did not exhibit neutrophil infiltration into the intestines (3). An intuitive approach would be to treat with antibiotics to reduce the

Significance

Our previous work has shown that bacterial-induced activation of neutrophil granulocytes after allogeneic hematopoietic cell transplantation promotes acute graft-versus-host disease (GVHD). In patients, depleting neutrophils is not possible because they mediate protection from invading bacteria and antibiotic treatment needed to reduce bacteria-induced neutrophil activation affects protective intestinal microbiota. Here, we describe that active and passive immunization against the conserved microbial surface polysaccharide poly-*N*-acetylglucosamine leads to killing of invading bacteria, which reduces the uncontrolled neutrophil activation, and thereby opens a new avenue to interfere with acute GVHD without affecting commensal intestinal microbial diversity.

Author contributions: J.H., C.K., B.B., G.B.P., G.H., and R.Z. designed research; J.H., O.S.T., E.H., S.U., N.G.N., S.T., Z.G., S.D., C.C.-B., O.O., S.K., A.S.-G., O.P., C.K., P.A., and M.J.B. performed research; B.B. and G.B.P. contributed new reagents/analytic tools; J.H., O.S.T., E.H., S.U., N.G.N., S.T., Z.G., S.D., C.C.-B., O.O., S.K., A.S.-G., O.P., C.K., J.D., P.A., M.J.B., B.B., G.B.P., G.H., and R.Z. analyzed data; and J.H., J.D., G.B.P., G.H., and R.Z. wrote the paper.

Conflict of interest statement: G.B.P. is an inventor of intellectual properties (human monoclonal antibody to poly-*N*-acetylglucosamine [PNAG] and PNAG vaccines) that are licensed by Brigham and Women's Hospital to Alopexx Vaccine, LLC, and OneBiopharma, Inc., entities in which G.B.P. also holds equity. As an inventor of intellectual properties, G.B.P. also has the right to receive a share of licensing-related income (royalties, fees) through Brigham and Women's Hospital from OneBiopharma, Inc., and Alopexx Vaccine, LLC. G.B.P.'s interests were reviewed and are managed by the Brigham and Women's Hospital and Partners Healthcare in accordance with their conflict of interest policies. C.C.-B. is an inventor of intellectual properties (use of human monoclonal antibody to PNAG and use of PNAG vaccines) that are licensed by Brigham and Women's Hospital to OneBiopharma, Inc. As an inventor of intellectual properties, C.C.-B. also has the right to receive a share of licensing-related income (royalties, fees) through Brigham and Women's Hospital from OneBiopharma, Inc.

This article is a PNAS Direct Submission.

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This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1908549116/-DCSupplemental.

First Published September 16, 2019.

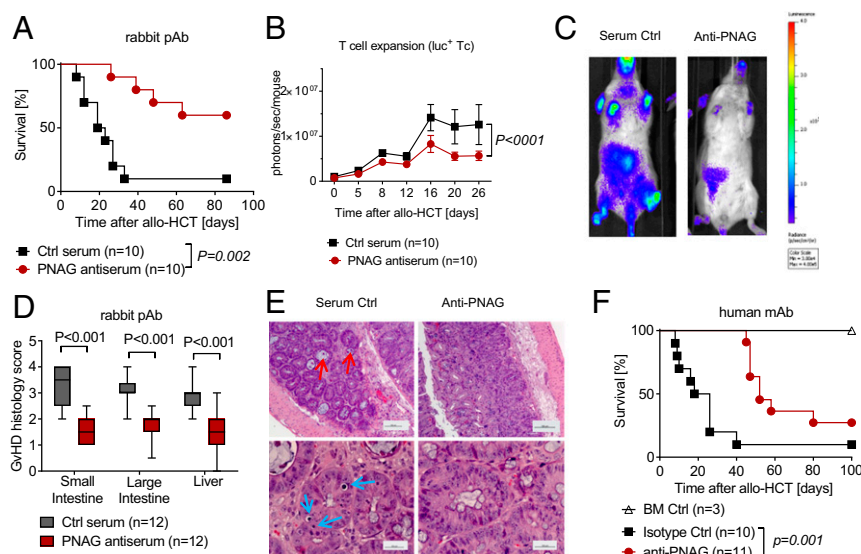


Fig. 1. Passive immunization against PNAG reduces GVHD mortality and severity. (A) Survival curves show recipient mice receiving either intraperitoneal (i.p.) polyclonal rabbit PNAG antiserum (rabbit pAb) or normal serum treatment as a control (Ctrl) on days 0, 3, 6 and 9. Data are pooled from 2 independent experiments. P values were calculated using the 2-sided Mantel–Cox test. (B and C) Bioluminescence imaging using i.p. injection of β -luciferin to detect expansion of Luc⁺ T cells (Luc⁺ Tc) during GVHD following allo-HCT in recipients from A. The P values were calculated by repeated-measures ANOVA using the area under the curve. Missing values were set to the mean value of remaining mice $[\text{Mean}(\text{Ctrl}) + \text{Mean}(\text{anti-PNAG})]/2$. For experiments shown in B, C Luc⁺ T cells were used. Representative images for each group are shown in C. (D and E) Histopathological scoring of GVHD severity of the small intestine, large intestine, and liver on day 8 following allo-HCT. Data are pooled from 2 independent experiments. The P values were calculated using the Mann–Whitney U test. The lines represent the medians, the upper and lower limits of the box plot represent the 25th and 75th percentiles, and the error bars depict the 10th and 90th percentiles. Representative images for each group are shown in E. The red arrows indicate crypt abscesses, and the blue arrows indicate apoptotic cells. (F) Survival curve of BALB/c recipient mice receiving either human monoclonal IgG1 anti-PNAG or isotype control monoclonal antibody (mAb) treatment (100 μg per dose) on days 0, 3, 6, and 9. BM control animals received only C57BL/6 bone marrow without T cells and were treated with isotype antibody (100 μg). Data are pooled from 2 independent experiments. P values were calculated using the 2-sided Mantel–Cox test.

invading bacteria. However, studies in mice showed that treatment with ampicillin, which affects Lactobacillales that otherwise expand during GVHD, causes more severe GVHD (5). Also, studies in mice and humans indicate that a decrease in microbial diversity, which often is a result of antibiotic treatment, is associated with an increased GVHD rate (5–7).

In clinical practice after allo-HCT, the use of antibiotics is often inevitable when patients are neutropenic; therefore, it would be desirable to have novel strategies that target invading bacteria without induction of massive changes in the diversity of the microbiota and, at the same time, reduce activation of neutrophils.

Poly-*N*-acetylglucosamine (PNAG) is a polysaccharide expressed on the outer surface of over 30 pathogens (8). This broad expression makes it a potential target for antibody treatment and vaccination, especially in settings where specific pathogens driving inflammation and pathology are either unknown or variable. Five to 15% of the PNAG monomer units are deacetylated and indispensable for the pathogenicity of some microbes (9). However, natural (10, 11) and vaccine-induced (12, 13) antibodies to the highly acetylated glycoform are poorly protective, unable to activate complement, and thus fail to mediate protective immunity. In contrast, antibodies raised against a highly deacetylated glycoform of PNAG (dPNAG) were effective at mediating opsonization, microbial killing, and protective immunity in multiple animal studies, including lethal peritonitis, bacteremia, and experimental colitis (8, 14, 15).

In contrast to treatment with antibiotics, which eliminates all sensitive bacteria, anti-PNAG treatment affects only bacteria that invade the intestinal submucosa as a consequence of conditioning-induced tissue damage and come into contact with effectors of adaptive immunity, including complement and phagocytes. Luminal bacteria most likely remain unaffected due

to insufficient levels of inflammatory cofactors such as cytokines, phagocytes, and complement in the luminal mucosal space.

In this study, we used a mouse model to investigate the impact of passively administered antibodies to PNAG and active vaccination against PNAG on the severity of GVHD. Sequencing analysis of the microbiota before and after allo-HCT showed that anti-PNAG treatment did not affect luminal microbial diversity. Mice receiving anti-PNAG treatment in the early phase after allo-HCT exhibited improved survival and reduced GVHD severity. Neutrophils recruited to the intestinal tract showed reduced myeloperoxidase activity in anti-PNAG-treated mice. Thus, we present an antimicrobial strategy that was less disruptive to the microbiota than antibiotic treatment, led to reduced neutrophil activation, and had protective effects in the setting of GVHD.

Materials and Methods

Mice. C57BL/6 (H-2Kb, CD45.1, or CD45.2) and BALB/c (H-2Kd or CD45.2) mice were purchased from Charles River Laboratories, Janvier Labs, or the local stock of the animal facility at the University of Freiburg. *LysM-Cre; Mcl1-*fl/fl** mice have been previously described (16). Mice were used between 6 and 14 wk of age, and only gender-matched transplantations were performed. Animal protocols were approved by the Regierungspräsidium Freiburg (no. G-18/036).

All other methods (blood and marrow transplant [BMT] models, bacterial vaccination, histopathology scoring, opsonic killing assays, enzyme-linked immunosorbent assay, sequencing, and sequencing data analysis) are described in *SI Appendix, Suppl. Methods*.

Statistical Analysis. Differences in animal survival (Kaplan–Meier survival curves) were analyzed by the Mantel–Cox test. To obtain unbiased data, a pathologist blinded to both the genotype and the treatment group performed the histopathological scoring of GVHD severity. For statistical analysis, an unpaired t test (2-sided) was applied. Data are presented as mean and SEM (error bars). If the data did not meet the criteria of normality, the Mann–Whitney U test was applied. For data analyzed by the nonparametric Mann–Whitney U test, the graphs show medians and a relevant range like

the 10th and 90th percentiles. Differences were considered significant when the *P* value was <0.05.

Results

aGVHD Severity Is Reduced by Anti-PNAG Treatment. Since microbial translocation to the gastrointestinal (GI) submucosa was previously shown to enhance aGVHD (17) and mice that lack innate immune activation receptors or downstream pathway effectors (18) exhibit less intestinal GVHD, we first tested the effect of a polyclonal rabbit anti-PNAG antibody (anti-PNAG antiserum) for its impact on mice developing aGVHD. We postulated that the antibody would impact inflammation and tissue destruction driven by bacteria in the GI submucosa and lessen GVHD-associated lethality. Groups of mice treated with the PNAG antiserum experienced significantly improved survival compared with mice treated with control serum (Fig. 1A). Notably, only 1 of 10 mice treated with PNAG antiserum died by day 30 after allo-HCT compared with 9 of 10 controls, and this occurred under experimental conditions wherein the last passive antibody infusion was given on day 9 after allo-HCT. Expansion of luciferase transgenic T cells was reduced in mice treated with the anti-PNAG antiserum compared with mice treated with control serum (Fig. 1B and C). In agreement with the improved survival, the histopathology scores for the small intestine, large intestine, and liver were lower in mice treated with the PNAG antiserum compared with mice treated with control serum (Fig. 1D and E). To test the general role of antimicrobial immunoglobulin G (IgG) in GVHD in a different model system, mice were immunized by subcutaneous injections with inactivated commensal bacteria and their serum was used to treat BMT recipients in a GVHD prevention approach. Survival of mice receiving injections of serum from mice immunized against commensal bacteria was improved compared with mice receiving control serum (SI Appendix, Suppl. Fig. S1A), supporting the concept that antimicrobial antibodies reduce GVHD-related death.

To further validate the potential efficacy of anti-PNAG passive immunotherapy, we tested a second approach by treating mice undergoing allo-HCT with the fully human IgG1 monoclonal antibody to PNAG (clone F598). Again, we observed improved survival of mice treated with the anti-PNAG antibody compared with mice treated with the isotype control (Fig. 1F). Similar to the results with rabbit PNAG antiserum, all mice treated with the monoclonal antibody to PNAG were alive at day 50 after allo-HCT, whereas 9 of 10 controls had died by day 40, and passive therapy was completed on day 9 after allo-HCT.

Anti-PNAG Treatment Does Not Affect Microbial Diversity, Reduces Intestinal Neutrophil Activation, and Induces Opsonic Killing Antibody to a Variety of Microbes. To test the hypothesis that anti-PNAG treatment was not impacting the diversity of the intestinal microbiota, we compared the diversity of the intestinal microbiome in mice treated with anti-PNAG after allo-HCT with that in mice receiving control antibody. The microbiota of mice treated with anti-PNAG antiserum did not show significant differences compared with the microbiota of serum control-treated mice before and on day 13 post allo-HCT, with the exception of selection for increased representation of the normally present *Bifidobacterium* species (Fig. 2A–C). It is known that opsonization of bacteria by antibodies and components of the complement system can lead to opsonophagocytic killing by macrophages and neutrophils, suggesting that the effect of anti-PNAG was to reduce microbial extraintestinal burdens and ameliorate the inflammatory response stemming from ineffective phagocytosis. In support of this hypothesis, we observed increased recruitment of neutrophil granulocytes to the ileum after total body irradiation (TBI) in anti-PNAG-treated mice (SI Appendix, Suppl. Fig. S1D). Consistent with a more rapid elimination of the

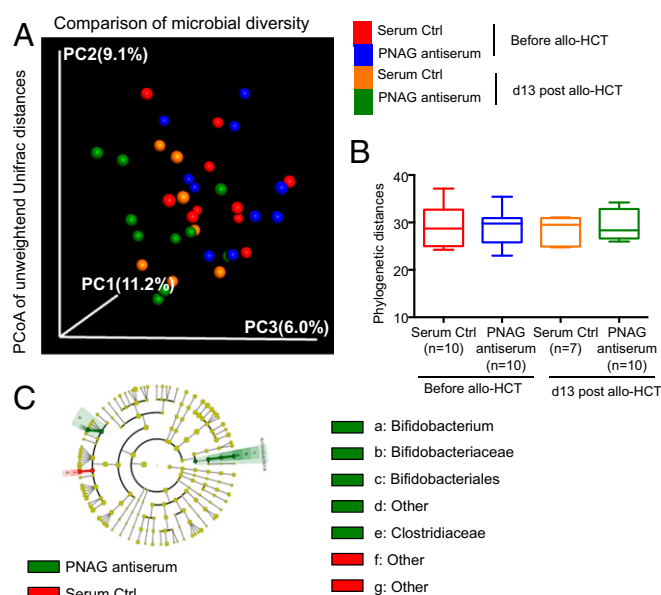


Fig. 2. Passive immunization against PNAG does not disturb intestinal microbial diversity. Comparison of the diversity of the microbiota in stool samples from allo-HCT BALB/c mice at a sequencing depth of 14,000 reads. (A) Principal component analysis (PCoA) of the unweighted Unifrac (a distance metric used to compare biological communities) distances. Significant differences were found before allo-HCT and on day 13 after allo-HCT in both control serum- and anti-PNAG serum-treated samples by the Adonis test ($P = 0.001$). PC, principal component. (B) Phylogenetic distance index presented by box plots (depth: 14,000 reads). No significant difference was found by the Mann-Whitney *U* test with 1,000 Monte Carlo simulations ($P > 0.05$). A comparison of microbiota changes in stool samples was analyzed by LEfSe (linear discriminant analysis effect size). The lines represent the medians, the upper and lower limits of the box plot represent the 25th and 75th percentiles, and the error bars depict the 10th and 90th percentiles. Ctrl, control. (C) Cladogram indicating the phylogenetic distribution of microbial lineages associated with clinical status; lineages with linear discrimination analysis (LDA) effect size ≥ 2.0 are displayed. Differences are represented in the color of the most abundant class (yellow: nonsignificant). Each circle's diameter is proportional to the taxon's abundance. Circles represent phylogenetic levels from domain to genus inside out. Control serum (red, $n = 7$) versus anti-PNAG (green, $n = 10$) on day 13.

bacteria, we observed decreased activity of neutrophils in anti-PNAG-treated mice by bioluminescence imaging when monitoring their myeloperoxidase activity, which is a major indicator for the activation of neutrophils (Fig. 3A–C). We then characterized the leukocyte composition in the intestinal wall in detail. Based on the expression of defined lineage markers, we identified major immune cell subsets in the myeloid (Fig. 3D and SI Appendix, Suppl. Fig. S2D) and T cell (SI Appendix, Suppl. Fig. S2A–C) compartments. Myeloid cells were analyzed for multiple markers, including MPO, Ki67, PD-L1, and others (Fig. 3D and SI Appendix, Suppl. Fig. S2D). We observed lower Ki67 expression within neutrophils in the anti-PNAG-treated mice compared with the serum control group (Fig. 3E and F). We further tested the hypothesis that neutrophils mediate the anti-PNAG effect using mice genetically modified to be devoid of neutrophils (*LysM^{cre} Mcl1^{fl/fl}*). No survival advantage was observed when *LysM^{cre} Mcl1^{fl/fl}* mice were treated with anti-PNAG compared with control serum (Fig. 3G). To validate this in a second setting, we generated a bone marrow (BM) chimera with *LysM^{cre} Mcl1^{fl/fl}* BM by syngeneic transplantation (*LysM^{cre} Mcl1^{fl/fl}*) into wild-type C57BL/6). The resulting chimera lacked neutrophils in the BM compartment and then underwent allo-HCT. In agreement with a role for neutrophils in the anti-PNAG-mediated effects, we observed no survival difference when recipients lacked neutrophils (Fig. 3H).

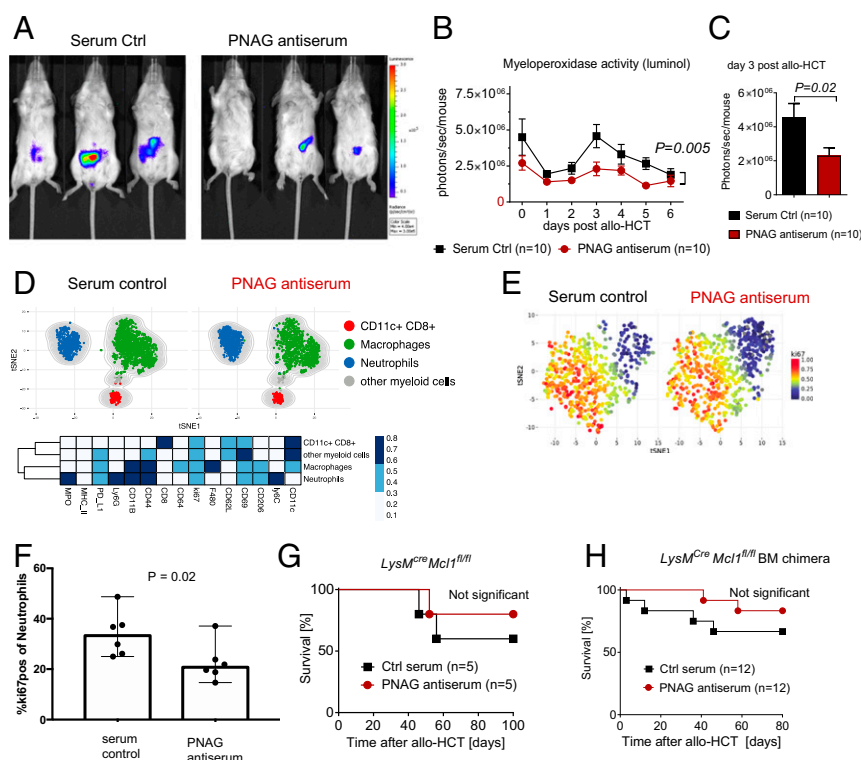


Fig. 3. Neutrophils contribute to anti-PNAG effect. Representative MPO activity images (A) and pooled data over time (B) are shown. Allo-HCT of C57BL/6 bone marrow cells into BALB/c recipient mice in combination with either polyclonal rabbit PNAG antiserum (anti-PNAG) or normal control (Ctrl) serum. Data are pooled from 2 independent experiments ($n = 10$). The P value was calculated using the area under the curve. (C) Bar diagram showing MPO signals of individual mice on day 3, the time point when MPO activity peaks. The P value was calculated using the 2-sided Student's unpaired t test. (D) Ileum of mice treated as described in A and B was isolated on day 2 after allo-HCT, and the myeloid compartment (gated on live/single cells/CD45⁺/CD11b or CD11c⁺/TCR- β cells) was analyzed. A t-distributed stochastic neighbor embedding (tSNE) plot with flow or mass cytometry data using a self-organizing map (FlowSOM)-guided metaclustering displays all cells from the 2 individual conditions, and the heat map shows the median marker expression (value range: 0 to 1) for each annotated population ($n = 6$ per group). The tSNE plot displays Ki67 expression of stochastically selected cells from neutrophils (Ly6G⁺/Ly6C⁺) from the 2 individual conditions (E) and frequency of manually gated neutrophils (live/single cells/CD45⁺/CD11b or CD11c⁺/TCR- β /Ly6G⁺/Ly6C⁺) expressing Ki67 (F). Bars in plot represent the median, and error bars depict the 95% confidence interval. The P value was calculated using the Mann-Whitney U test. (G) *LysM^{Cre} Mcl1^{fl/fl}* C57BL/6 mice were used as recipients for allo-HCT from BALB/c donors. Recipient mice received either human monoclonal anti-PNAG or isotype control (100 μ g) intraperitoneally (i.p.) on days 0, 3, 6 and 9. (H) *LysM^{Cre} Mcl1^{fl/fl}* C57BL/6 chimeric mice were used as recipients for allo-HCT from BALB/c donors. Recipient mice received either human monoclonal anti-PNAG or isotype control (100 μ g) i.p. on days 0, 3, 6 and 9.

Since neutrophils are highly sensitive to irradiation, they will rapidly disappear after TBI with 1,110 cGy and will not be detectable in the murine terminal ileum after 1 wk, as previously reported by us (3, 4). Consistently, we found hardly any neutrophils on day 8 after TBI in the terminal ileum, while on day 2 after TBI, neutrophils were detectable (SI Appendix, Suppl. Fig. S1 B and C). We therefore chose to analyze cellular responses on day 2 or 3 after allo-HCT. We also characterized the T cell compartment in the intestine in the anti-PNAG-treated group compared with the serum control-treated group (Fig. 4A and B). We observed increased frequencies of effector memory CD4 T cells (T_{EM} cells) in the anti-PNAG group compared with the serum control group (Fig. 4C). The increase in T_{EM} cells was connected to lower GVHD activity in studies reported by others (19); however, T_{EM} cells were of donor origin in these reports, while in our analysis on day 2, the T cells were mainly of recipient origin. Consistent with reduced immune activation, interleukin-12 was reduced in the serum of anti-PNAG-treated mice compared with control mice (Fig. 4D).

These results are most likely explained by the effective opsonic killing of extraintestinal microbes, leading to resolution of inflammatory responses due to lower microbial burdens. Previous studies (8) have shown opsonic killing activity of polyclonal anti-PNAG animal sera against multiple microbes.

Here, we confirmed that this was also a property of the sera used in this study, using the fully human IgG1 monoclonal antibody to PNAG as a control and reference reagent (Fig. 5A) and sera obtained from dPNAG-vaccinated mice and humans (Fig. 5B).

In total, all of these findings support the concept that neutrophils are less activated when microbes can be eliminated by anti-PNAG antibody treatment.

Vaccination-Induced Anti-PNAG Titers Persist after Allo-HCT and Reduce Disease Severity. Vaccination of mice using a penta-saccharide fragment of PNAG (5GlcNH₂) coupled to tetanus toxoid (TT) (20) (Fig. 6A) induced increased anti-PNAG antibody titers compared with mice treated with the solvent control (SI Appendix, Suppl. Fig. S3C). In mice vaccinated against PNAG, high anti-PNAG antibody titers were still found on day 8 after allo-HCT (Fig. 6B). Mice vaccinated against PNAG exhibited less GVHD-related death compared with mice treated with solvent control (Fig. 6C). The presence of the antibodies until 8 d after allo-HCT was sufficient to confer the beneficial effect observed for passive immunization. In further support of the efficacy of PNAG vaccination, we tested whether human immune serum derived from subjects in a clinical trial evaluating the safety and immunogenicity of the 5GlcNH₂-TT vaccine (ClinicalTrials.gov identifier: NCT02853617) could show an effect on survival after passive transfer to mice. A pool of the anti-PNAG

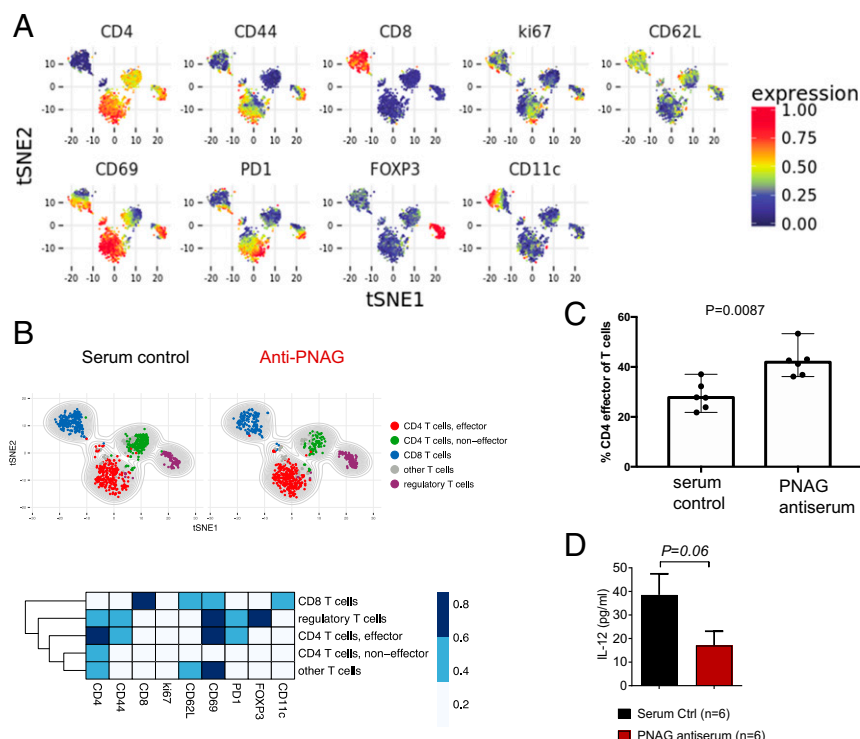


Fig. 4. Anti-PNAG treatment affects the effector CD4 T cell compartment. (A and B) BALB/c mice were treated with either polyclonal rabbit PNAG antiserum or control serum (200 μ L) and underwent allo-HCT. As shown in A, the ileum was isolated on day 2 after allo-HCT and analyzed for T cell markers as indicated (gated on live/single cells/CD45⁺/F4/80⁺TCR- β ⁺). Flow or mass cytometry data using a self-organizing map (FlowSOM)-guided metaclustering displays all cells from the 2 individual conditions ($n = 6$ per group). tSNE, t-distributed stochastic neighbor embedding. (B and C) Percentage of CD4⁺ effector T cells within the T cell population is shown for the serum control and anti-PNAG groups ($n = 6$ per group). Bars in the plot represent the median, and error bars depict the 95% confidence interval. The P value was calculated using the Mann-Whitney U test. (D) Serum level of interleukin-12 (IL-12) is shown for the serum control (Ctrl) and anti-PNAG groups ($n = 6$ per group). The P value was calculated using the 2-sided Student's unpaired t test.

immune human serum obtained after vaccination reduced GVHD-related death in the aGVHD mouse model compared with human serum obtained from the same individuals before vaccination (Fig. 6C). Overall, these findings indicate that the antimicrobial activity of the vaccination against PNAG-expressing bacteria reduces GVHD severity.

Discussion

aGVHD is often preceded by viral (21, 22) or bacterial (23) infections that could serve as an initial trigger for the latter allogeneic immune response. For many years, the intuitive strategy was to use antibiotic prophylaxis, which was associated with lower GVHD risk and death in initial studies (20). However, antibiotic treatment also leads to a loss of intestinal microbial diversity (5) and, in several studies, to a higher incidence of GVHD (7, 24). Therefore, strategies that allow for elimination of invading bacteria, while sparing luminal commensal bacteria, could help to reduce GVHD.

Approaches that rely on the elimination of bacteria by antibodies and additional immune effectors could have an advantage over antibiotics because immunotherapies would mainly target bacteria that have translocated into the recipient's tissues, particularly the intestinal submucosa or subdermal layers of the skin. At these sites, PNAG-expressing microbes could be recognized by antibody, complement, and innate immune cells and could be eliminated. Since PNAG is expressed on the surface of over 30 pathogens (8), immunotherapies targeting this antigen could cover a broad spectrum of microbes that invade tissues below body surfaces during the early phase of tissue damage following allo-HCT. It is likely that some of the bacteria, which are not targeted by the anti-PNAG antibody, cause tissue damage and

promote GVHD. However, the relative reduction of bacteria bound by anti-PNAG antibody was sufficient to reduce GVHD and improve survival in the majority of the animals. In line with the concept that anti-PNAG did not eliminate all invading bacteria, GVHD was not fully blocked by the anti-PNAG treatment as we still detected minor histopathological GVHD signs in the anti-PNAG group, as shown in Fig. 1D. Examples of bacteria that are eliminated by anti-PNAG treatment and were previously reported to be associated with GVHD include *Enterococcus*, *Fusobacteria*, and *Akkermansia*, as well as the fungi *Candida* and *Aspergillus* (7, 25–28).

Our results are consistent with previous studies that have shown anti-PNAG antibodies are effective at opsonizing multiple microbial strains, leading to opsonophagocytic or bactericidal killing. This translated into better outcomes in preclinical models of infection with multiple pathogens (8). In this report, we extend these pathogen-specific protective activities to show that either active immunization of mice with a PNAG-targeting vaccine or passive treatment with rabbit PNAG antiserum, a fully human monoclonal antibody, or polyclonal human anti-PNAG antiserum, when administered in the early phase after allo-HCT, reduced aGVHD-related mortality and histopathological severity. This is in agreement with a study showing reduction of GVHD severity when mice were treated with chicken antibodies directed against multiple intestinal pathogens (29). We extend this study by using different antibodies directed against a specific bacterial antigen (PNAG) that has advanced to phase 1 human testing and has been tested in patients (ClinicalTrials.gov identifier: NCT02853617), and we show that active immunization against PNAG targets bacteria that invade the recipient's tissues, rather than targeting luminal bacteria in the intestinal tract.

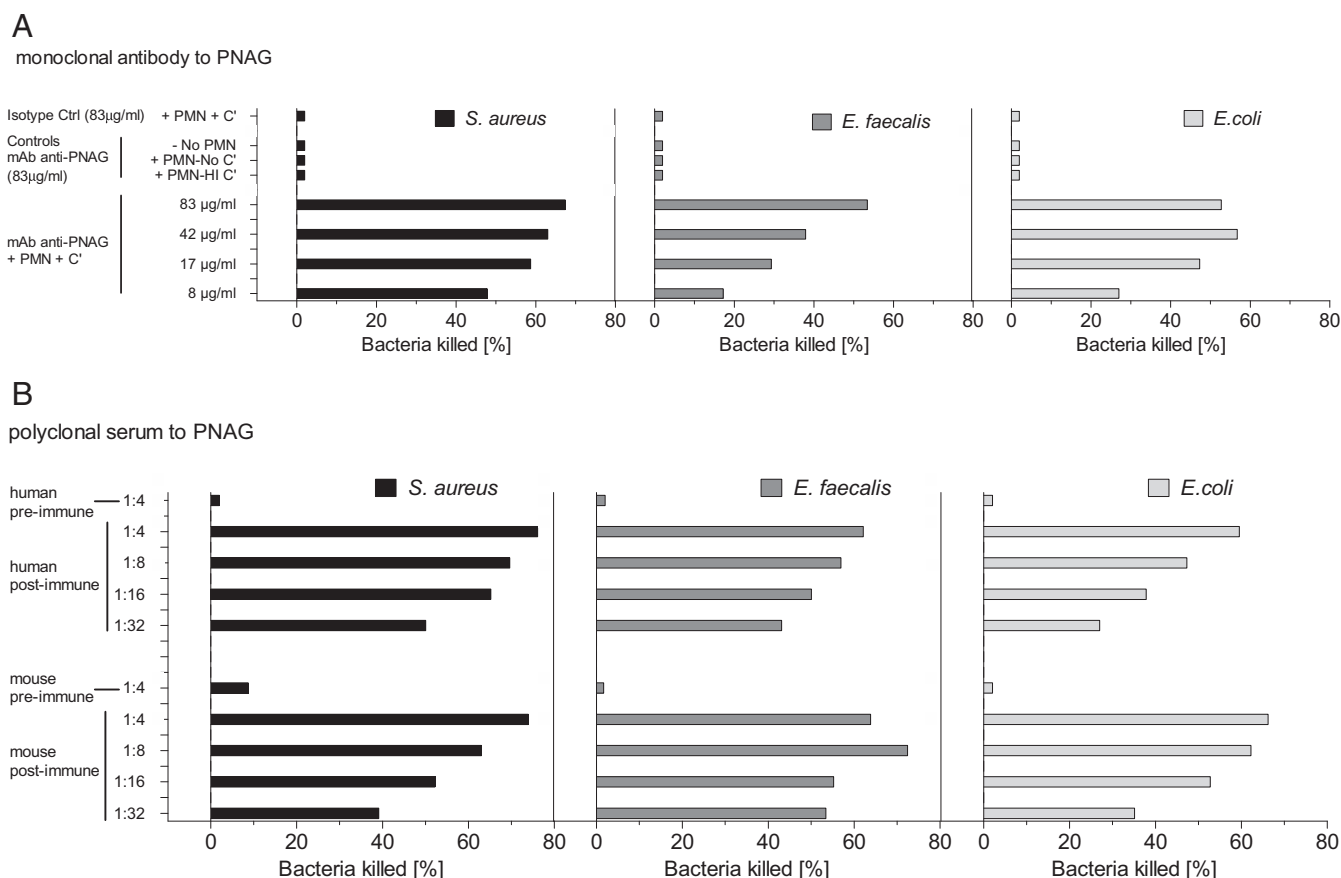


Fig. 5. Polyclonal sera and human monoclonal antibody to PNAG induce opsonic killing against a variety of microbes. Opsonic killing activity of human monoclonal antibody (mAb) to PNAG (F598) or in sera raised to 5GlcNH₂-TT anti-PNAG vaccine in the presence of neutrophils (PMN) and complement (C'). (A) Killing of *Staphylococcus aureus* (strain MN8), *Enterococcus faecalis* (strain V583), or *Escherichia coli* (K1 strain E11) at different PNAG antibody concentrations compared with controls. Isotype Ctrl, IgG1 control; No PMN, absence of neutrophils; No C', absence of complement; HI C', heat-inactivated complement. (B) Killing of *S. aureus*, *E. faecalis*, and *E. coli* with antisera from humans and mice raised to 5GlcNH₂-TT (postimmune) at different serum dilutions compared with sera obtained before immunization (preimmune).

Importantly, we observed that anti-PNAG antibody treatment did not reduce intestinal microbial diversity as determined by sequencing analysis of 16S ribosomal DNA. Mechanistically, we could

show that anti-PNAG antibody treatment caused more abundant neutrophil recruitment to the intestinal submucosa. This supports the concept that neutrophils are able to effectively eliminate any

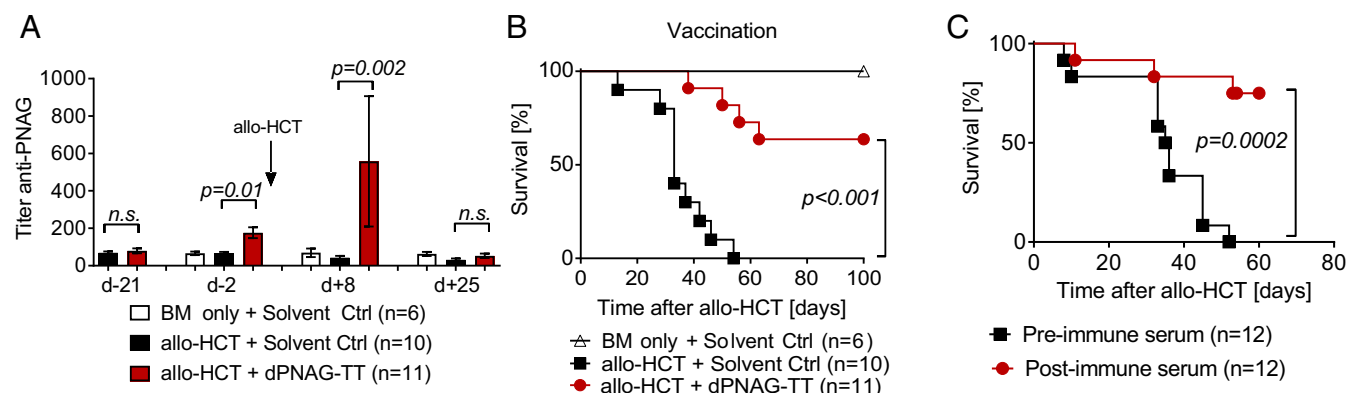


Fig. 6. Active immunization against PNAG raises antibody titers against PNAG and reduces GVHD mortality. (A) Anti-PNAG titers are shown as determined at different time points by enzyme-linked immunosorbent assay. On day -21, TT-only control mice are combined. The *P* values were calculated using the 2-sided Student's unpaired *t* test. Ctrl, control; n.s., not significant. (B) Survival of allo-HCT recipient Balb/C mice immunized with either 5GlcNH₂-TT or TT-only as described in A. The *P* value was calculated using the 2-sided Mantel-Cox test. (C) BALB/c mice received allo-HCT from C57BL/6 donor mice (5×10^6 BM and 3×10^5 CD4⁺/CD8⁺ T cells). Recipient mice received either pre- or postimmune human serum intraperitoneally on days 0, 3, 6, and 9. Human serum was derived from a pool of 3 individuals that received 2 doses of 150 µg in 0.2% alum 28 d apart from the 5GlcNH₂-TT vaccine. Preimmune serum was obtained from the same individuals before vaccination, and postimmune serum was obtained 28 d after the second vaccination. Data are pooled from 2 independent experiments. The *P* value was calculated using the 2-sided Mantel-Cox test.

opsonized bacteria in this tissue, leading to reduced local inflammation. Tissue-resident myeloid cells may recognize opsonized bacteria, which then may be a signal to attract neutrophils. In agreement with this proposal, we observed lower intestinal inflammation, reduced myeloperoxidase activity, and proliferation of neutrophils in the mice treated with anti-PNAG antibody.

In summary, we demonstrate that the novel properties of anti-PNAG antibody treatment and vaccination against PNAG reduced GVHD-related mortality. This could be translated into a clinical application, given the modest toxicity profile of vaccination and the availability of fully human anti-PNAG antibodies, both of which have been tested in phase 1 trials in humans ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study?term=NCT02853617) identifier: NCT02853617). Mechanistically, anti-PNAG antibodies enhance antimicrobial immunity while reducing the activation of

neutrophil-mediated downstream immunopathology, which is a promising approach to reduce the severity of GVHD.

ACKNOWLEDGMENTS. This study was supported by Deutsche Forschungsgemeinschaft (DFG) TRR167 Project B06 (to R.Z.) and SFB1160 TP Project B09 (to R.Z.); European Research Council (ERC) Consolidator Grant 681012 GvHDCure (to R.Z.); Deutsche Krebsstiftung Grant 111639 (to R.Z. and G.H.); an Excellence Initiative of the DFG (GSC-4, Spemann Graduate School), (CIBSS-EXC 2189), and Deutsche Gesellschaft für Mukosale Immunologie und Mikrobiom (to C.K.); and the C&D Fund (to M.J.B.). O.O. was supported by the Tinsel Cultural Foundation and Istanbul University. C.C.-B. was supported by an unrestricted gift from Alopexx Vaccine, LLC. This work was supported by grants from the Swiss Cancer League (to B.B.); Swiss National Science Foundation (310030_170320, 316030_150768, and CRSII5_183478) (to B.B.); European Union FP7 Project Advanced T-cell Engineered for Cancer Therapy (ATECT) (to B.B.); and the University Research Priority Project Translational Cancer Research (to B.B. and N.G.N.).

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